

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MAR 321 — Past Questions & Materials

by Renance

Course: MAR 321

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 3

Levels: 300

This study pack was compiled and restructured by Renance for MAR 321 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — MAR 321

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: Second Semester

SESSION: 2024/2025

DATE: 02/10/2025



COURSE CODE: MAR 321

COURSE TITLE: Navigation & Meteorology

TIME ALLOWED: 3 Hrs

INSTRUCTION: ATTEMPT ANY 5 QUESTIONS

Question 1

- i. One of the Essential components of a ship is the Navigational Equipment. Describe the types of ship Navigational Equipment studied. (8 marks)
- ii. As a Seaman you are expected to understand the various emergency sounds onboard ship. What are the following alarms and explain the duties to do on hearing the alarms
 - [a] General Emergency Alarm
 - [b] Fire Alarm
 - [c] Man Overboard Alarm
 (6 marks)

Question 2

- i. There are various terminologies used onboard ship as regards to Navigation. Explain the following terminologies;
 - [a] Bridge [b] Navigation [c] Meteorology [d] Seamanship [e] Forecastle [f] MCR
 (6 marks)
- ii. Navigational Aids are very important to all Seafarers. Discuss what you understand by ship Navigational Aids and generate examples (8 marks)

Question 3

- i. Anchor is very important to ship, as it secures a ship from drifting. Suggest the types and sub-types of the recently designed anchors used onboard ship. (5 marks)
- ii. Two points $P(32^{\circ}N, 47^{\circ}W)$ and $Q(32^{\circ}N, 28^{\circ}E)$ are on the sea surface. Calculate
 - (a) The radius of the parallel of latitude through P
 - (b) The distance between P and Q along the parallel of latitude
 - (c) If it takes a Tanker vessel 9 hours to sail from P to Q, calculate the speed of the vessel in knot (9 marks)

Question 4

- i. For effective movement of a ship from one point to another, the Captain and other deck officers must be very competent in Nautical Measurements. Briefly discuss what is meant by "Nautical miles" in marine operations (3.5 marks)
- ii. A cargo-passenger vessel is 2 nautical miles away from ashore. Calculate the distance of the vessel in (a) cable length (b) feet (c) meters (d) statute miles (e) fathom (f) league (10.5 marks)

Question 5

- i. Meteorology as a branch of Maritime studies is a prerequisite for all Marine Engineers. Explain the Parts of the Atmosphere. (2.5 marks)
- ii. Explain Climatology and its branches (5.5 marks)
- ii. Discuss the Meteorological Equipment Studied (6 marks)

Question 6

- i. For Easy hoisting, transferring and lowering of Cargoes and other operations onboard ship, of Deck machinery and Fitting are used. Explain the Classification of Deck machinery and Fitting, stating the function or application of each of them. (10 marks)
- ii. Packaged Cargoes are mostly transported in various Container Units. Suggest the Types of Container Units used for such purpose. (4 marks)

Question 7

- i. With the aid of a chart, Describe the organizational arrangement of crew member in a merchant ship, ranging from the Captain to the least crew member (5 marks)
- ii. It is mandatory by IMO that every Seafarer must undergo basic safety training to enable them withstand emergency situations onboard ships. Suggest the types of Safety Equipment used onboard marine vessels (4 marks)
- iii. Generate a brief summary of accessing a scene where accident has just occurred onboard a ship. (6 marks)

GOOD LUCK

Paper 2 of 3 — MAR 321

Level 300 • Second Semester • PastQuestion

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COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: Second Semester

SESSION: 2024/2025

DATE: 02/10/2025



COURSE CODE: MAR 321

COURSE TITLE: Navigation & Meteorology

TIME ALLOWED: 3 Hrs

INSTRUCTION: ATTEMPT ANY 5 QUESTIONS

Question 1

- i. One of the Essential components of a ship is the Navigational Equipment. Describe the types of ship Navigational Equipment studied. (8 marks)
- ii. As a Seaman you are expected to understand the various emergency sounds onboard ship. What are the following alarms and explain the duties to do on hearing the alarms
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 - [b] Fire Alarm
 - [c] Man Overboard Alarm
 (6 marks)

Question 2

- i. There are various terminologies used onboard ship as regards to Navigation. Explain the following terminologies;
 - [a] Bridge [b] Navigation [c] Meteorology [d] Seamanship [e] Forecastle [f] MCR
 (6 marks)
- ii. Navigational Aids are very important to all Seafarers. Discuss what you understand by ship Navigational Aids and generate examples (8 marks)

Question 3

- i. Anchor is very important to ship, as it secures a ship from drifting. Suggest the types and sub-types of the recently designed anchors used onboard ship. (5 marks)
- ii. Two points $P(32^{\circ}N, 47^{\circ}W)$ and $Q(32^{\circ}N, 28^{\circ}E)$ are on the sea surface. Calculate
 - (a) The radius of the parallel of latitude through P
 - (b) The distance between P and Q along the parallel of latitude
 - (c) If it takes a Tanker vessel 9 hours to sail from P to Q, calculate the speed of the vessel in knot (9 marks)

Question 4

- i. For effective movement of a ship from one point to another, the Captain and other deck officers must be very competent in Nautical Measurements. Briefly discuss what is meant by "Nautical miles" in marine operations (3.5 marks)
- ii. A cargo-passenger vessel is 2 nautical miles away from ashore. Calculate the distance of the vessel in (a) cable length (b) feet (c) meters (d) statute miles (e) fathom (f) league (10.5 marks)

Question 5

- i. Meteorology as a branch of Maritime studies is a prerequisite for all Marine Engineers. Explain the Parts of the Atmosphere. (2.5 marks)
- ii. Explain Climatology and its branches (5.5 marks)
- ii. Discuss the Meteorological Equipment Studied (6 marks)

Question 6

- i. For Easy hoisting, transferring and lowering of Cargoes and other operations onboard ship, of Deck machinery and Fitting are used. Explain the Classification of Deck machinery and Fitting, stating the function or application of each of them. (10 marks)
- ii. Packaged Cargoes are mostly transported in various Container Units. Suggest the Types of Container Units used for such purpose. (4 marks)

Question 7

- i. With the aid of a chart, Describe the organizational arrangement of crew member in a merchant ship, ranging from the Captain to the least crew member (5 marks)
- ii. It is mandatory by IMO that every Seafarer must undergo basic safety training to enable them withstand emergency situations onboard ships. Suggest the types of Safety Equipment used onboard marine vessels (4 marks)
- iii. Generate a brief summary of accessing a scene where accident has just occurred onboard a ship. (6 marks)

GOOD LUCK

Paper 3 of 3 — MAR 321 Navigation & Meteorology 2020 2021 Second Semester

Level 300 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF TECHNOLOGY
DEPARTMENT OF MARINE ENGINEERING

EXAMINATION: Second Semester

SESSION: 2019/2020

DATE: 13 /07/2021



COURSE CODE: MAR 321

COURSE TITLE: Navigation & Meteorology

TIME ALLOWED: 2 Hrs :30 mins

INSTRUCTION: ATTEMPT ANY 5 QUESTIONS

Question 1

- i. Define or describe the following terminologies;
 [a] Seamanship [b] Navigation [c] Meteorology. [d] Catamaran [e] Bilge keel [f] Funnel (6 marks)
- ii. Discuss what you understand by ship Navigational Aids. (8 marks)

Question 2

- i. What are the Two (2) basic distress alarm bells on-board marine vessels, Describe their sounds (4 marks)
- ii. Describe the types of ship Navigational Equipment studied. (10 marks)

Question 3

- i. In the International ship code of signaling and communication, what are the words given to Alphabets A to Z and their individual meanings (9 marks)
- ii. With the aid of a chart, Describe the organizational arrangement of crew member in a merchant ship, ranging from the Captain to the least crew member (5 marks)

Question 4

- i. What is Deck Machinery (2 marks)
- ii. Discuss the Classification of Deck machinery and Fitting, stating the function or application of each of them. (12 marks)

Question 5

- i. What is the meaning of the following terminologies in ship Cargo operations
 [a] Dunnage [b] Hatch Coanings [c] RO-PAX. [d] Lockups [e] Cargo Liner [f] Collision Bulkhead (6 marks)
- ii. Explain the ways you can enhance the security or protection of your ship as a crew member. (9 marks)

Question 6

- i. With the aid of a well illustrative diagram, describe a ship Anchor (5 marks)
- ii. State the similarities and Dissimilarities of Anchoring and Mooring (4 marks)
- iii. List the types and sub-types of the recently designed anchors studied. (5 marks)

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